Artavazd Maranjyan

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Google Scholar Linkedin

Current Position

2026–present **Postdoctoral Researcher, working with Volkan Cevher and Martin Jaggi**, *EPFL*, Lausanne, Switzerland

Education

2023–2025 **Ph.D. in Computer Science**, *KAUST*, Thuwal, Saudi Arabia; **Advisor:** Peter Richtárik
Thesis: First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data

2021–2023 **M.Sc. in Applied Statistics and Data Science**, *Yerevan State University*, Yerevan, Armenia
Thesis: On local training methods; **Co-supervisors:** Peter Richtárik, Mher Safaryan

2017–2021 **B.Sc. in Informatics and Applied Mathematics**, *Yerevan State University*, Yerevan, Armenia
Thesis: On the Convergence of Series in Classical Systems; **Supervisor:** Martin Grigoryan

Selected Industry and Academic Experience

03/2025 **Research Visit to Yi-Shuai Niu**, *BIMSA*, Beijing, China
- Gave talks at three universities (PKU, BUAA, BIMSA)
- Collaborated with Professor Yi-Shuai Niu on a project on Server-Assisted Federated Learning

06/22–01/23 **Internship in the Group of Peter Richtárik**, *KAUST*, Thuwal, Saudi Arabia
- Worked on the "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity" paper

07/21–06/22 **Co-Founder**, *OnePick*, Yerevan, Armenia
- Built an MVP platform for generating or selecting photos matching text descriptions
- Participated in the InVent 2.0 (FAST) venture building program, from idea generation to pitching
- Secured initial funding and continued development beyond the program

09/19–01/21 **Software Engineer in Test**, *Picsart*, Yerevan, Armenia
- Worked with automation team to design and develop automated solutions across several mobile/web applications
- Worked with software developers, test engineers, product owners, business analysts to find and resolve issues
- Worked closely with DevOps to suggest improvements in processes and in Jenkins Continuous Integration cycle

Awards

06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowships**, *EPFL*
Competitive fellowship supporting independent postdoctoral research, with additional travel funding

05/2025 **CEMSE Dean's List Award**, *KAUST*
Awarded for excellent academic and research performance (USD 2,500)

09/2023 **Dean's Award**, *KAUST*
Awarded to a few top students accepted to KAUST (USD 18,000 over three years)

05/2021 **Outstanding Final Project Award**, *Yerevan State University*
Recognized for the Bachelor's thesis (awarded to 6 students among 250+ students)

Selected Publications

7. **Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method**
Abdurakhmon Sadiev, Artavazd Maranjyan, Ivan Ilin, Peter Richtárik
arXiv:2605.18174, 2026

6. **Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity**
Ammar Mahran, Artavazd Maranjyan, Peter Richtárik
arXiv:2605.13434, 2026

5. **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity**
Artavazd Maranjyan, Peter Richtárik
***ICLR** 2026: The Fourteenth International Conference on Learning Representations*

4. **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity**
Artavazd Maranjyan, Alexander Tyurin, Peter Richtárik
ICML *2025: Forty-Second International Conference on Machine Learning*
3. **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning**
Artavazd Maranjyan, El Mehdi Saad, Peter Richtárik, Francesco Orabona
ICML *2025: Forty-Second International Conference on Machine Learning*
2. **MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
Artavazd Maranjyan, Omar Shaikh Omar, Peter Richtárik
UAI *2025: The 41st Conference on Uncertainty in Artificial Intelligence*
OPT 2024: Optimization for Machine Learning (NeurIPS workshop)
Oral presentation (top 5% of 107 submissions)
1. **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity [video]**
Artavazd Maranjyan, Mher Safaryan, Peter Richtárik
TMLR 2025: Transactions on Machine Learning Research

Reviewing

2024–present TMLR (Transactions on Machine Learning Research)
2026 NeurIPS (Conference on Neural Information Processing Systems)
2026 ICML (International Conference on Machine Learning)
2025 ICLR (International Conference on Learning Representations)
2024 SIMODS (SIAM Journal on Mathematics of Data Science)
2024 JMLR (Journal of Machine Learning Research)

Selected Talks

26/04/2026 ***Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)***, *Lagune Barra Hotel*, Rio de Janeiro, Brazil
Delivered a talk titled **Controlling Delay in Asynchronous SGD: Optimality for Any Data Regime** [video] [slides]

29/07/2025 ***35º Colóquio Brasileiro de Matemática***, *IMPA*, Rio de Janeiro, Brazil
Delivered a talk and a poster on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides] [poster]

16/04/2025 ***FLOW Talk #126***, *Federated Learning One World Seminar (FLOW)*, Online
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [video] [slides]

17/03/2025 ***Academic Report of the School of Mathematical Sciences***, *Beihang University (BUAA)*, Beijing, China
Invited by Jiaxin Xie to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]

13/03/2025 ***Optimization Seminar***, *BIMSA*, Beijing, China
Invited by Yi-Shuai Niu to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]

12/03/2025 ***Seminar***, *Peking University*, Beijing, China
Invited by Kun Yuan to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]

21/11/2024 **MLR Weekly Seminar**, *Machine Learning Research at Apple*, Online
Invited by Samy Bengio to give a talk on **MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [slides]

13/03/2024 ***The Machine Learning Summer School in Okinawa 2024***, *OIST*, Okinawa, Japan
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]

06/10/2023 ***Algorithms & Computationally Intensive Inference seminars***, *University of Warwick*, Coventry, England
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [slides]